

Response to Reviewers

Manuscript (revised): *Do Cryptocurrency Markets Differentiate Infrastructure from Regulatory Shocks? A Multi-Moment Event Study with Dependence-Robust Inference* **Original title:** *Infrastructure vs Regulatory Shocks: Asymmetric Volatility Response in Cryptocurrency Markets*
Journal: *Digital Finance* (Springer) **Author:** Murad Farzulla (Dissensus) **Decision:** Major Revision

Overview: this revision overturns its own headline, on purpose

I want to set out the shape of this revision before the point-by-point response, because it is larger than the decision letter requested and it changes the paper's central claim.

The original submission's headline was that infrastructure events generate a **5.7× larger conditional-variance response** than regulatory events, with $p < 0.001$. Reviewer 2's central comment (R2.1) asked whether that multiplier survives when the event-selection screen is relaxed. Answering it properly required me to reconstruct the candidate pool and re-run the inference end to end — and that process surfaced two defects in the inference behind the headline. Neither is a data problem; neither changes the point estimate. Both are errors in how significance was computed, and once corrected, **the headline does not survive**. The asymmetry is directional but not statistically distinguishable from zero under dependence-robust inference.

Rather than patch the original claim, I have rebuilt the paper around the correction. The revision does four things:

1. **Reports a dual null at both moments, honestly.** I test the infrastructure-regulatory difference at the **first moment (returns)** and the **second moment (conditional variance)** on one shared sample under one matched dependence-robust design. Returns: +7.19 pp, block-bootstrap $p = 0.283$, Ibragimov–Müller $p = 0.295$. Variance: directional $\approx 2.6\text{--}5\times$, Student- t -copula bootstrap $p = 0.322$ (baseline), 0.318 (crisis control). Neither survives.
2. **Makes the inference cautionary tale the lead contribution.** A new section (§*Why the Asymmetry Was an Artefact*) sets out four inference lessons — pseudoreplication, the concentrating nature of cross-asset dependence, heavy-tailed bootstrap innovations, and block-versus-parametric resampling at the first moment — as a self-correction. This is framed as methodological maturity, not self-flagellation: I followed the correct inference and it dissolved my own result. The section is now anchored by two new pieces of apparatus: an **inference ladder** (a single table walking the *identical* comparison through seven progressively-more-correct inference procedures, from the dead headline $t = 4.768 / p = 0.0008$ down to the inference of record $p = 0.322$, with the p -value rising monotonically as each defect is corrected), and a **Monte-Carlo size study** (described below) that shows the over-rejection is a general property of the naive procedure, not a feature of this one dataset.

3. **Subsumes the companion returns analysis.** The first-moment (CAR) analysis previously circulated as a separate companion (the *no-structure / Same Returns, Different Risks* paper) is folded into this manuscript, so the two moments are tested and reported together rather than across two papers.
4. **Keeps the two findings that survive** as ballast: the scope condition (curated vs mechanical event identification, R2.1) and the weekly Granger result (R2.3).

What I am explicitly *not* claiming: any significant asymmetry at either moment; that the effects are zero or the shock types identical (the first moment is underpowered — MDE $\approx 19\text{--}20$ pp vs observed ≈ 7 pp — so this is a failure to reject); and a 5%-significance claim anywhere in the paper.

Disclosure: the two inference defects behind the original $p = 0.0008 / 5.7\times$

I flag these directly so the reviewers do not have to reconstruct them by comparing rounds.

Defect 1 — pseudoreplication. The original significance test ($t = 4.768$, $p = 0.0008$) treated the six per-asset GJR-GARCH-X coefficients as six independent observations. They are not: all six assets see the *same* 50 events and correlate 0.54–0.83 (mean $\bar{\rho} \approx 0.69$), so the effective sample size is far below six. The coefficients cluster tightly because they average the same correlated events, not because the effect is precisely estimated. Event-level re-tests (the unit of independence) collapse it: Welch $p = 0.52$, Mann-Whitney $p = 0.54$, cluster-robust panel (SE clustered by event) $p = 0.50$. The model-free proxies are low-powered and bracket the truth from below; the uncorrected test brackets it from far above. A design-effect correction on the actual estimator (effective $N \approx 1.4$) lands the honest value at $t \approx 1.8\text{--}2.3$, $p \approx \mathbf{0.07}$ — marginal at 10%, never the 5% the uncorrected test implied.

Defect 2 — Gaussian bootstrap innovations under heavy tails. Moving to a cross-asset CCC-GARCH-X bootstrap addressed the dependence but left a second defect: it drew standardised innovations from a **Gaussian** while each model is fitted with **Student-*t*** errors at $\nu \approx 3$. With $\nu \approx 3$ the true innovation is far heavier-tailed, so the Gaussian draw mis-specifies the null distribution and biases the bootstrap p downward. Correcting the draw to the fitted Student- t margins (a multivariate- t copula, per-asset ν , rescaled to unit variance) moves the baseline p from **0.059 (Gaussian) to 0.322 (Student-*t*)**.

Together, the two corrections convert the original $p \approx 0.001$ into a failure to reject, with the point estimate ($\approx 5\times$) unchanged at every stage. Each correction is implemented in replication code (c5/c6 pseudoreplication re-tests; c7 Gaussian CCC bootstrap; c9 the corrected Student- t -copula bootstrap that is the inference of record).

A Monte-Carlo size study certifies the diagnosis (new since the prior round)

A reviewer could reasonably ask whether I have simply walked one dataset from a low p to a high one and called it a correction. To pre-empt that, I added a Monte-Carlo **size-distortion study** (c10) that turns the diagnosis from an assertion into a measured fact. I fit a true-null DGP to the data — per-asset GJR-GARCH-X estimated under a null-imposed *combined* event dummy (so $\delta_{\text{infra}} = \delta_{\text{reg}}$ by construction; there is genuinely no differential effect in the truth), cross-asset dependence at $\bar{\rho}_z = 0.705$, Student- t margins at the fitted ν (median 3.82) with joint tail

dependence via a true multivariate- t copula — then simulate 1,000 panels (923 usable) and measure how often each inference procedure rejects a true null.

The result is exactly the demonstration the paper needs:

Inference procedure	Size @ 0.05	Size @ 0.10
(i) Naive i.i.d. t -test	0.512 $\pm$ 0.016	0.701 $\pm$ 0.015
(ii) Design-effect corrected	0.434 $\pm$ 0.016	0.655 $\pm$ 0.016
(iii) Gaussian-copula bootstrap	0.308 $\pm$ 0.015	0.398 $\pm$ 0.016
(iv) Student- t -copula bootstrap	0.039 $\pm$ 0.006	0.095 $\pm$ 0.010

The naive i.i.d. rule — the one that produced the headline $t = 4.768 / p = 0.0008$ — rejects a true null **51% of the time at the nominal 5% level** ($\approx 10\times$ nominal) and 70% at 10%. The design-effect fix helps but under-corrects; the Gaussian-copula bootstrap (right dependence, wrong tails) improves but stays over-sized at 31%; only the Student- t -copula bootstrap restores nominal size (3.9% / 9.5%). Crucially, the t -copula rung is **self-validating**: it is calibrated from the same DGP the panels are drawn from, so its size *must* land near nominal up to Monte-Carlo error — and it does, which certifies the simulation has no bug. The headline $p = 0.0008$ was, with high probability, one of the 51% of true nulls the naive rule falsely rejects. This study is new since the prior round and is reported as §A *Monte-Carlo size study* with its full DGP in an appendix.

Reviewer 1

“Overall, this is a strong and carefully executed manuscript that fits well within the scope of *Digital Finance*.”

The reviewer’s comments largely concerned tone, framing, and presentation. Most are not merely satisfied but superseded by the reframing — the revised paper is, by construction, the moderate-claim version Reviewer 1 was steering toward.

R1.1 — Tone and strength of causal language

Comment: “Some statements adopt very strong causal or normative language. A slight moderation would improve credibility.”

Response: This is now resolved at the level of the conclusion, not just the wording. The paper no longer asserts a significant asymmetry to soften — it reports a dual failure to reject. Every causal and significance claim attached to the headline has been removed or downgraded. The abstract, introduction, results, discussion, and conclusion state the result as “we cannot reject equality of infrastructure and regulatory event impacts at either moment, on this sample, under correct inference,” with the underpowered caveat stated symmetrically throughout. There is no 5%-significance claim anywhere in the manuscript.

R1.2 — Event classification and endogeneity (exogeneity framing)

Comment: *“Explicitly framing infrastructure events as closer to exogenous shocks would strengthen identification.”*

Response: Retained and still relevant. Infrastructure failures (unannounced outages, externally-discovered exploits, banking-partner insolvencies) arrive at calendar times essentially exogenous to current market conditions; regulatory timing is closer to endogenous (political cycles, prior market behaviour, leaks). The manuscript states this asymmetric identification quality explicitly. Note that under the new framing this strengthens the honesty of the null rather than the strength of a claim: even granting infrastructure events their cleaner exogeneity, the difference is not statistically resolvable on this sample.

R1.3 — Length and focus

Comment: *“Some robustness material could be moved to an online appendix to improve readability.”*

Response: The revised manuscript is reorganised around the dual-null result and the four inference lessons, which are now the spine of the paper rather than appended robustness. Secondary robustness (event-window sensitivity, placebo, temporal stability, Bayesian validation, the anticipation and winsorisation checks) is reported in compact form with headline numbers in the body and full per-asset tables in the replication materials. I am happy to relocate any block the editor prefers to an online appendix.

R1.4 — Network analysis as a secondary contribution

Comment: *“The ETH versus BTC centrality result ... should be framed more clearly as a complementary finding.”*

Response: Resolved by demotion, consistent with the prior revision and the new framing. Eigenvector centrality on a six-node correlation graph is weakly discriminating (ETH 1.00 vs BTC 0.97; all six between 0.85 and 1.00) and is reported as a supplementary observation only, not a headline contribution. It carries no causal or systemic-hub claim.

R1 Minor — Overlapping event windows

Comment: *“Clarify handling of overlapping event windows.”*

Response: The overlap-handling protocol is stated explicitly with the affected pairs and the weighting decision in each case. This matters more under the new framing, since the shared 50-event sample is now used at both moments and the overlap treatment must be consistent across them; it is.

R1 Minor — Near-integrated volatility persistence

Comment: *“Briefly discuss near-integrated volatility persistence.”*

Response: Addressed jointly with R2.2. Bai-Perron-style (PELT) break tests detect 2–4 breaks per asset clustering around the November 2022 FTX collapse; within-sub-sample persistence is

on average lower than the full-sample $\alpha + \beta \approx 0.988$ (sub-sample mean ≈ 0.90), though within the wide estimation noise of the short regimes. The persistence drop is reported as suggestive, not significant (no sub-segment drop exceeds two standard errors) — in keeping with the paper’s general discipline of not over-reading.

R1 Minor — Positioning within the digital finance literature

Comment: “Strengthen positioning within the digital finance literature.”

Response: The literature section now positions the contribution against operational-resilience and microstructure work and, more centrally, against the methodological literature on pseudoreplication, few-cluster inference (Ibragimov–Müller), and heavy-tailed bootstrap calibration — which is where the paper’s lead contribution now sits.

Reviewer 2

“The economic significance of the findings is well-established... the author implements several robustness checks.”

Reviewer 2’s R2.1 (selection bias) is the comment that set this entire revision in motion. It receives the fullest response.

R2.1 — Selection bias from the “demonstrable market-wide impact” criterion

Comment: (Summarised) *The Stage-2 criterion (≥ 2 assets showing market-wide impact) risks selecting on the dependent variable; infrastructure failures almost always clear it while regulatory news may be systematically rejected, biasing the comparison toward H1. The reviewer asks for (a) a drop-out rate by event category and (b) a sensitivity analysis at a relaxed threshold.*

Response: This comment is the proximate cause of the whole reckoning, and it is answered on its own terms — by *measuring* selection bias rather than arguing it away — and then carried further than the reviewer asked, into the inference correction.

(a) Drop-out census. I reconstructed a systematic candidate pool of **135 events** (82 infrastructure, 53 regulatory, Jan 2019–Aug 2025) from public-record sources (Rekt/DeFiLlama hack leaderboards; SEC/CFTC enforcement records; major-jurisdiction regulatory announcements; exchange status pages) and applied the *identical* Stage-2 impact filter to both legs. Pass rates:

Category	N	Stage-2 pass rate
Infrastructure	82	64.6%
Regulatory	53	75.5%

The filter is **less** demanding for regulatory candidates than for infrastructure candidates (two-proportion $z = -1.33$, n.s.) — the *opposite* sign to the selection-bias hypothesis. The screen does not mechanically favour infrastructure. Because the same filter is applied to both source-defined

populations, the comparison inherits no asymmetric selection from how the pool was assembled, even though the reconstructed 135-pool is not the original (unpreserved) 208-item working list.

(b) Relaxed-threshold sensitivity — the scope condition. Re-estimating the GJR-GARCH-X specification under a properly-nested mechanical sweep on the reconstructed pool:

Spec	N infra	N reg	Multiplier	Welch p
Curated sample (full 4-stage protocol)	26	24	4.88× (5.7× canonical)	0.0015
No filter (135)	82	53	0.58×	0.446
≥1 asset (115)	69	46	1.49×	0.321
≥2 assets (93, like-for-like)	53	40	1.61×	0.164
≥3 assets (77)	42	35	1.31×	0.403

On the broad pool the multiplier is a modest **1.3–1.6×**, **not significant at any threshold**; the like-for-like two-asset screen (the mechanical analogue of the curated impact criterion) gives 1.61×, nowhere near 5.7×. The strong magnitude is therefore a property of **curated, high-salience event identification**, not of a mechanical impact threshold. The ~3× curated-vs-mechanical gap reproduces under a consistent canonical winsorisation (curated 5.24× vs two-asset 1.63×), so it is an identification effect, not a winsorisation artefact.

Correction relative to the prior revision draft (disclosed). An intermediate draft mis-framed the sweep as “peaking at the two-asset threshold,” placing the curated 50-event sample at the “two-asset point.” That was wrong on two counts: the curated sample is selected by the full four-stage protocol, not a mechanical two-asset screen, and its count (50) is smaller than the three-asset screen (77), so the counts cannot nest. The genuine two-asset screen on the reconstructed pool (93 events) gives 1.61× ($p = 0.16$), and the result is now presented as a scope condition, not a peak.

How R2.1 escalated into the inference correction. Re-running the estimation to answer R2.1 is what exposed that the curated-sample significance itself (not just the magnitude) did not survive proper inference. The curated 5.7× / $p < 0.001$ was a pseudoreplication-plus-Gaussian-bootstrap artefact (see Overview, Defects 1–2). So the selection-bias answer has two layers: (i) the magnitude is identification-dependent (scope condition), and (ii) even the curated-sample significance evaporates under dependence-robust inference (t -copula $p = 0.322$). Both are reported. The scope condition is retained as a substantive finding; it is no longer load-bearing for a significance claim, because there is none.

R2.2 — Bai-Perron structural-break tests

Comment: “*The high persistence parameter often masks structural breaks ... perform formal tests for multiple structural changes (Bai and Perron, 1998).*”

Response: Implemented and retained, and expanded into a full control battery. A PELT-based multiple-break procedure (BIC-penalised, trimming 0.15, max 5 breaks, bootstrap CIs on break

locations) detects 2–4 breaks per asset, clustering at the November 2022 FTX collapse; within-sub-sample persistence falls from a full-sample mean of $\alpha + \beta \approx 0.985$ to ≈ 0.905 . I am now explicit that this drop is **descriptive only** — when each sub-segment’s persistence is estimated with its own standard error, *none* of the 27 sub-segments falls more than two SEs below its full-sample value (the regimes are short, SEs ± 0.06 – 0.31), and the drop is not a small-sample artefact (mean persistence 0.891 for $n \geq 500$ segments vs 0.887 for $n < 500$). The near-integration framing is therefore qualified, not overturned, and reported as suggestive.

The break regimes also serve the second-moment inference directly, and the revision now reports a fuller break-control battery with its **cross-asset-robust** significance (the prior round’s break-control p -values were naive Welch tests — the same pseudoreplication the headline suffered). Under the copula bootstrap: adding a single FTX-crisis-regime control moves the multiplier only from $4.88\times$ to $3.97\times$ and the robust p from 0.323 to 0.318 (a real chunk of the magnitude is crisis-baseline variance, but the differential is not *merely* the crisis regime); the full-regime control halves the multiplier to $2.64\times$ ($p = 0.249$, reliability-flagged at 23% degenerate refits) and inverts XRP alone. The complete battery — break controls, anticipation sweep, winsorisation invariance, Ljung-Box/AR(1), and persistence SEs — is in §*Control battery for the second-moment point estimate*. A formal Markov regime-switching estimate of crisis amplification (H3) is motivated by the break evidence but left as future work; given the dual null it is moot for the headline and I prefer not to report an untested amplification.

Anticipation, disclosed as a new robustness exhibit. I also added an anticipation sweep: if regulatory events were anticipated, lengthening the regulatory pre-window would absorb leaked pre-event volatility into δ_{reg} and shrink the multiplier toward one. The opposite occurs — as the reg pre-window lengthens from 3 to 10 days, δ_{reg} *falls* ($0.405 \rightarrow 0.224$), the infrastructure coefficient is invariant, and the multiplier *grows* ($4.88\times \rightarrow 8.77\times$). Anticipation runs against the asymmetry, not for it. This is defensive ammunition for the point estimate, not a significance claim (which remains the copula bootstrap’s, and null).

R2.3 — Granger causality for sentiment and volatility

Comment: “*The relationship between news sentiment and volatility could be reflexive ... consider a Granger causality framework.*”

Response: Implemented, and it surfaced and corrected a methodological error of my own. GDELT sentiment is a **weekly** series. An initial pass ran the test at daily frequency by forward-filling the weekly values, which turns the predictor into a step function (constant within each week) that is mechanically incapable of carrying daily lead-lag information — the forward-fill was destroying the signal, and the resulting null was an artefact. Run correctly at the native weekly frequency, sentiment **Granger-leads** volatility for **10 of 18** asset-sentiment pairs, **7 surviving Benjamini–Hochberg at $q < 0.05$** (12 at $q < 0.10$). This is reported as the one genuinely positive empirical result in the paper, alongside the scope condition, and the daily $\rightarrow$ weekly correction is disclosed candidly.

R2.4 — Diebold-Mariano / out-of-sample forecasting discussion

Comment: “*A more thorough discussion ... to compare out-of-sample forecasting performance ... against a standard GARCH(1,1) benchmark.*”

Response (candid). As disclosed in the prior round, the out-of-sample DM figures carried in the original draft were not reproducible from the analysis pipeline, which implements in-sample model comparison. I report what the pipeline supports — GJR-GARCH-X attains the lowest AIC for four of six assets — and flag a rolling-window, HLN-corrected out-of-sample DM evaluation as future work. Under the new framing this is lower-stakes: the paper’s contribution is the dependence-robust inference on the event differential, not a forecasting-superiority claim, and I would rather under-claim than report a figure the code does not produce.

R2.5 — Correlation heatmap

Comment: “Consider adding a correlation matrix, perhaps as a heatmap.”

Response: Added (6×6, hierarchically clustered, annotated). The cross-asset correlation it displays (mean $\bar{\rho} \approx 0.69$, range 0.54–0.83) is no longer just descriptive — it is the quantity that drives the entire inference correction (the dependence that makes the six per-asset coefficients pseudoreplicated). The figure therefore does double duty: descriptive in §4.1, and the empirical premise of the four inference lessons. The data-vintage consistency fixes from the prior round (BTC-ETH 0.83; XRP range 0.54–0.65; XRP annualised volatility 77.3% as the maximum) are retained.

R2.6 — “Relative independence” wording for $\rho \approx 0.5$

Comment: “A correlation of this magnitude is a moderate-to-strong relationship, not ‘relative independence.’”

Response: Rephrased. XRP’s pairwise correlations (0.54–0.65) are described as moderate-to-strong but below the BTC-ETH benchmark. This is consistent with the paper’s central point: even the *least* correlated asset is correlated enough that treating the six as independent is pseudoreplication.

R2.7 — Constant-mean-return assumption made explicit

Comment: “Explicitly mention in Section 3.4 that the models assume a constant average return.”

Response: Done and justified empirically: $r_t = \mu + \varepsilon_t$ with constant μ ; Ljung-Box flags serial correlation in only 8/18 cells, fitted AR(1) coefficients are tiny, and an AR(1)-in-mean refit leaves the event coefficients essentially unchanged. (Under the new framing the point estimate’s stability matters more than its significance, and it is stable.)

R2.8 — Expanded parameter table

Comment: “Expand Table 2 to include α , β , θ_1 , θ_2 , and v .”

Response: Done. The full variance-equation parameter vector (ω , α , γ , β , v , δ_{infra} , δ_{reg} , θ_{reg} , θ_{infra}) is reported per asset. The fitted Student- t degrees of freedom ($v \approx 3$) are now load-bearing rather than incidental: they are exactly why a Gaussian bootstrap mis-specifies the null (Lesson 3). The small-negative GJR γ estimates are read against Baur and Dimpfl (2018)’s documented inverted crypto leverage effect.

γ (GJR leverage) column correction — carried forward

The data-integrity correction to the γ column disclosed in the prior round is retained: the original positive, starred γ values were stale carry-overs; the correct estimates are the small-negative values (-0.066 to -0.001), sourced directly from the model-output JSON, with the significance stars and the equity-exceeding-leverage claims removed and the negative γ reconciled with the literature (Baur and Dimpfl 2018).

On the title and scope change

The original title asserted an “asymmetric volatility response.” The revised title — *Do Cryptocurrency Markets Differentiate Infrastructure from Regulatory Shocks? A Multi-Moment Event Study with Dependence-Robust Inference* — poses the question the data can honestly address and signals both the multi-moment design and the inference focus. The paper now subsumes the companion first-moment (returns) analysis, so the question is answered once, at both moments, on one sample, under one design. I recognise that retracting my own headline against an open Major-Revision slot is unusual; I have made the change because correct inference required it, and I have written the manuscript to make the editors’ judgment on re-review straightforward.

Summary of changes

Area	Change	Origin
Title / framing	Retitled to the multi-moment question; dual failure-to-reject is now the conclusion; inference cautionary tale is the lead contribution	R2.1 → full inference reckoning
Headline	$5.7\times / p < 0.001$ retracted; replaced by directional-but-non-significant at both moments (returns $p = 0.283$; variance t -copula $p = 0.322$)	Defects 1–2
New first-moment analysis	CAR comparison (+7.19 pp) via event-level block bootstrap + Ibragimov–Müller; underpowered caveat (MDE ≈ 19 –20 pp); subsumes the <i>no-structure</i> companion	new + R1.1
New inference section	Four lessons: pseudoreplication, concentrating dependence,	Defects 1–2

Area	Change	Origin
	heavy-tailed bootstrap innovations, block vs parametric	
New: inference ladder	Single table walking the identical comparison through 7 inference rungs, $p = 0.0008 \rightarrow 0.322$ monotone	Defects 1–2
New: Monte-Carlo size study	True-null DGP; naive rule over-rejects 51% @ 5% ($\approx 10\times$ nominal); t -copula bootstrap restores nominal size (3.9%); self-validating	new (c10)
Inference of record (variance)	Student- t -copula CCC-GARCH-X bootstrap ($B = 2000$); replaces the Gaussian draw; Gaussian-vs- t comparison table across all 3 break specs	Defect 2
§Scope condition	Curated $5\times$ vs mechanical 1.3 –1.6 $\times$ (never significant); drop-out census (reg passes higher, 75.5% vs 64.6%)	R2.1
Expanded control battery	Break-regime table (robust p), anticipation sweep table ($4.88\times \rightarrow 8.77\times$), winsorisation invariance, Ljung-Box/AR(1), persistence SEs	R2.2
§Granger	Weekly sentiment-leads-volatility (10/18; 7 at FDR $q < 0.05$) with full per-asset table; daily forward-fill artefact disclosed	R2.3
Bai-Perron	Breaks at FTX; within-regime persistence ≈ 0.905 (descriptive only, 0/27 drops > 2 SE); crisis-control robust inference reported	R2.2, R1 minor
Fuller appendices	t -copula bootstrap algorithm (step-by-step), GARCH-X spec, Monte-Carlo DGP	new
Presentation	Heatmap (R2.5), full parameter table (R2.8),	R1.x, R2.x

Area	Change	Origin
	constant-mean justification (R2.7), “relative independence” rephrased (R2.6), network demoted (R1.4), exogeneity explicit (R1.2), overlap protocol (R1 minor)	
Corrections carried forward	γ column, units (variance not volatility), AIC 4/6, retracted Corbet (2019) reference removed	prior round
DM / out-of-sample	In-sample AIC reported; out-of-sample DM flagged as future work; unreproducible figures removed	R2.4

The paper is a more honest object than the original submission. The selection-bias comment that prompted this revision did exactly what good refereeing should: it pushed me to inspect the inference, and the inference did not hold. I have followed it where it led.